

SECS/GEM to Ignition

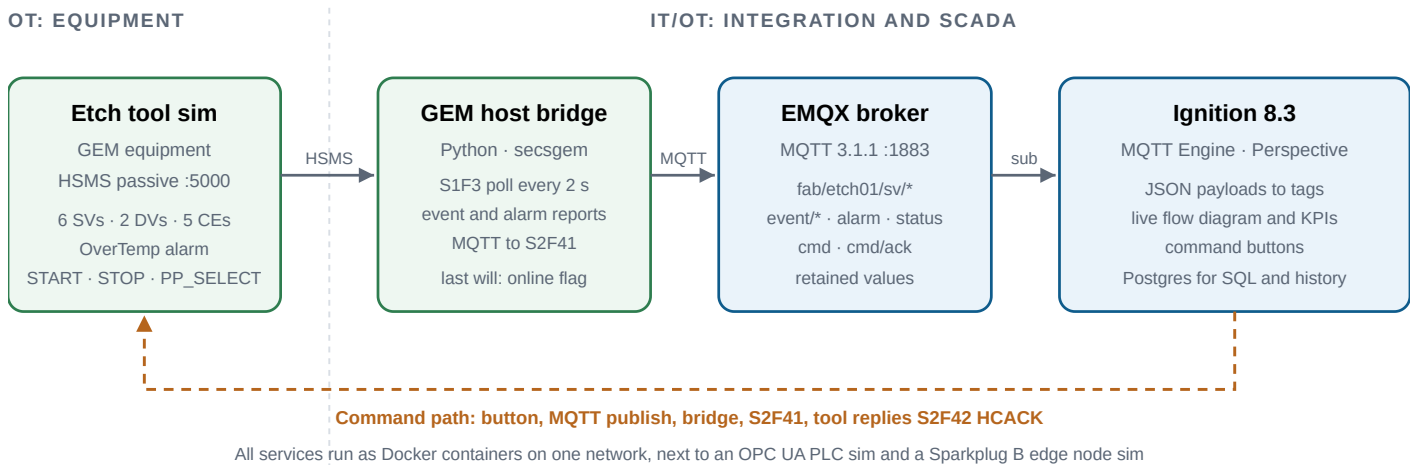
A closed-loop fab tool integration lab: equipment data up to SCADA, remote commands back down

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Home lab build, September 2026

This lab connects a simulated semiconductor etch tool to an Ignition 8.3 SCADA gateway using the same protocols a fab uses. The tool speaks SECS/GEM over HSMS. A Python GEM host collects its data and republishes it over MQTT, and Ignition turns it into live tags and an operator screen. Commands from the screen travel back down the same path and reach the tool as standard S2F41 remote commands.

Architecture



What it demonstrates

Equipment integration

- SECS/GEM host behavior: establish communication, subscribe to collection events, enable alarms, poll status variables
- Event reports carry context data (lot and wafer ID) with each WaferStart and WaferEnd
- Remote commands with parameters, acknowledged with HCKACK

IT/OT architecture

- A unified namespace pattern: one topic tree per tool under `fab/<tool>`
- Decoupled layers: the tool, bridge, broker and SCADA can each be swapped or scaled on their own
- Bridge health through an MQTT last will, shown live on the screen

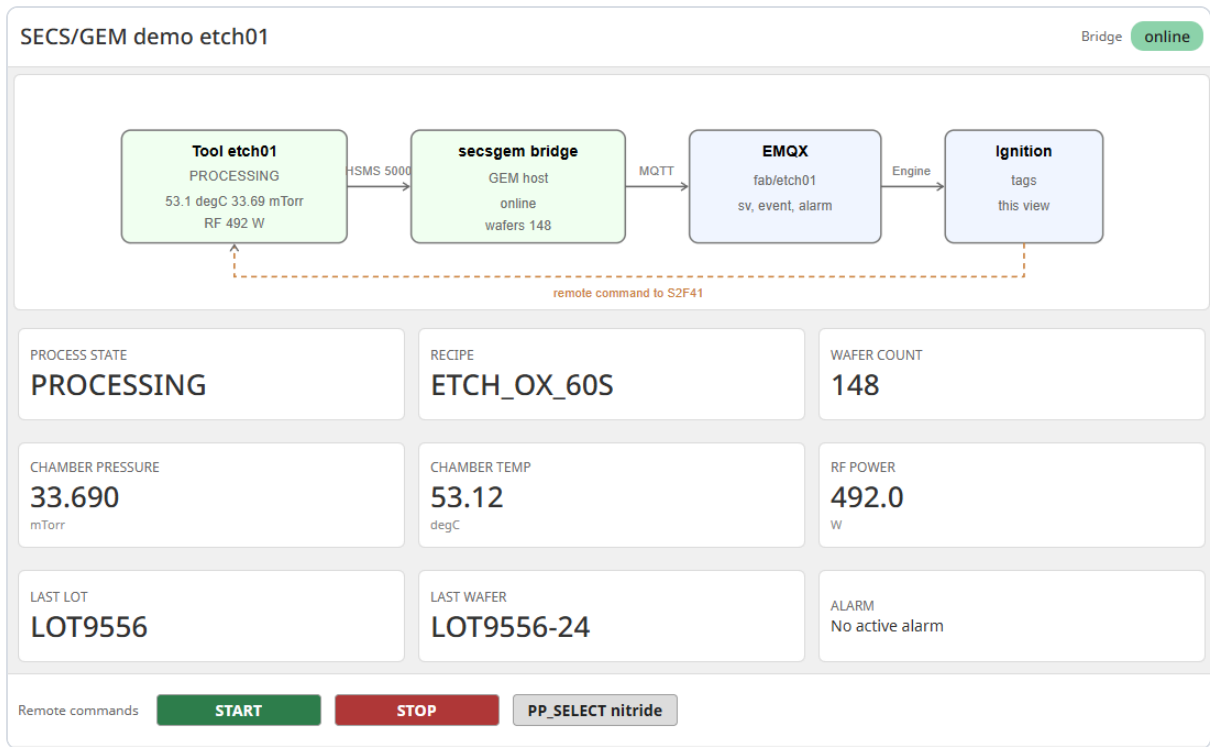
SCADA and HMI

- Ignition Perspective view with a live SVG flow diagram built from tag values
- Tiles for state, recipe, wafer count, chamber pressure, temperature and RF power
- Alarm tile that shows only active alarms

Engineering practice

- Everything defined as code: Docker Compose, Python, view JSON
- Repeatable import and backup scripts in PowerShell
- Issues traced end to end with logs and broker tools, documented on page 3

The operator screen



Live Ignition Perspective view. The flow diagram colors and values come from tags: the tool box turns green while PROCESSING, and the bridge box follows its online status. The buttons send remote commands to the tool.

A command round trip

- 1 The operator presses **PP_SELECT nitride**. A Perspective script publishes `{"rcmd": "PP_SELECT", "params": {"PPID": "ETCH_NIT_90S"}}}` to `fab/etch01/cmd` through MQTT Engine.
- 2 EMQX delivers it to the bridge, which is subscribed to the command topic.
- 3 The bridge sends **S2F41** Host Command Send to the tool over HSMS.
- 4 The tool replies **S2F42** with HCAK 4 (accepted, will finish later) and changes its recipe.
- 5 The bridge publishes the ack to `cmd/ack`. On the next S1F3 poll the new recipe shows on screen, within about 2 seconds.

Data model

Type	IDs	Items
SV	2101 to 2106	ChamberPressure, ChamberTemp, RFPower, ProcessState, WaferCount, Recipe
DV	2001, 2002	LotID, WaferID
CE	3001 to 3003	WaferStart, WaferEnd, RecipeSelected
CE	3101, 3102	OverTemp set and clear
ALID	5001	OverTemp, above 80 degC
RCMD		START, STOP, PP_SELECT(PPID)

MQTT topics under `fab/etch01`

<code>sv/<Name></code>	retained, <code>{"ts", "value"}</code>
<code>event/<Name></code>	CE with lot and wafer data
<code>alarm</code>	retained, set flag and text
<code>status</code>	online flag, last will
<code>cmd</code> , <code>cmd/ack</code>	remote command and HCAK

Build and debug notes

Getting the loop working end to end took five fixes. Each was found by checking one layer at a time: container network, broker, tag browser, binding preview, gateway console and bridge log.

Symptom	Cause	Fix
Bridge could not reach the broker	The add-on services joined Compose's default network, not the lab network that EMQX and Ignition use	Put every service on the shared <code>lab</code> network and made the bridge wait for EMQX to report healthy
Flow diagram showed ERROR, bridge showed null	The status payload is <code>{"online":true}</code> , so the tag is <code>status/online</code> . The view pointed at <code>status/online/value</code> , and one bad reference made the whole expression bad quality	Corrected the path in all four bindings
Alarm text stayed on after clearing	The retained alarm message carries text on both set and clear	Bound the tile to the <code>set</code> flag as well
Buttons said "sent" but the tool did nothing	The script published to a server named <code>emqx</code> , but MQTT Engine calls the connection <code>Chariot SCADA</code> . No error was raised	Moved the server name into one view property. Confirmed with the bridge log and a <code>mosquitto_sub</code> listener
Silent publish failures	The original scripts had no error handling	Byte payloads, try and except, and logging to the console and gateway log

Technology

SCADA	Ignition 8.3, Perspective, MQTT Engine
Equipment protocol	SECS-II, GEM, HSMS via Python secsgem 0.3
Messaging	EMQX, paho-mqtt, Sparkplug B sim
Other sources	OPC UA PLC simulator
Data	PostgreSQL 17
Platform	Docker Compose, PowerShell

How it scales to a fab

- One bridge per tool, or Ignition's own SECS/GEM module as host. A second sim tool on port 5001 is already set up for that
- Topic tree by site, area and tool, feeding MES, SPC and a historian from one broker
- Tag history on the process values for trends and FDC style limits
- TLS and per-client ACLs on the broker, command topics limited to approved roles
- Command audit trail from `cmd` and `cmd/ack` into SQL